TECHNICAL EVALUATION

Technical and security overview

Pilot / Early Access

Identity, permissions, retained technical evidence, provider boundaries and operating responsibilities for the scoped pilot.

Equipment. People. The work around them.

Asset and AssetLight provide the general foundation. Standard develops that foundation for industry-specific work; Metrology is the principal current edition.

Agree product configuration, evaluation scenarios and adoption terms before rollout.

PRODUCT FIT

Choose the scope around your work.

The products share a foundation. Specialization and the agreed operating scope determine the fit.

Asset / Pilot / Early Access	Agnostic item records, audit trails and general business management. Includes the AssetLight foundation, with customer work, purchasing, people and invoicing in its intended general business scope.
AssetLight / Pilot / Early Access	Focused instrumentation management. Equipment identity, location, service history and supporting records. Confirm the import and employee workflows needed for your pilot.
Standard by MetraSlate: Metrology Edition / Pilot / Early Access	Calibration laboratory workflows on the Asset foundation. CMC budgets, supported point uncertainty, calibration records, technical review, certificates and customer service.
Future editions	Standard Public Safety is in development. Standard Traceability is in design. Their publications describe intended direction.

Current scope

Agree the configuration, evaluation scenarios and adoption terms with MetraSlate before rollout. Package exposure and accepted customer workflows must be confirmed for the selected pilot.

SECURITY, IDENTITY AND ACCESS

Security belongs in the operating foundation.

Review organizational identity, company boundaries, application permissions and connection handling in the configuration you intend to use.

Pilot / Early Access. Identity and authorization depend on the deployment and configured roles. Architecture is not a certification, audit opinion or guarantee of compliance.

Organizational identity, company isolation, application permissions and protected connection credentials are part of the architecture. These mechanisms are material for a deployment-specific security review. No SOC 2 report, independent penetration-test result or security certification is claimed here.

WorkOS-backed identity connects the organization to supported sign-in and enterprise identity administration. Single Sign-On, directory synchronization and domain administration use the available setup and entitlements. The application resolves company membership and permissions alongside that identity, so signing in and authorizing a consequential business action remain separate decisions.

Company scope follows the request into data access. Tenant-scoped connections require an explicit company context, and database policies constrain company records. Application authorization distinguishes laboratory members, platform operators and portal customers, with additional role and privilege checks where the operation requires them.

Customer portal access follows the account and permitted sites. Internal conversations, customer-service conversations, released documents and technical approvals each have their own access rules. The person who may read a record does not automatically acquire permission to change, approve or release it.

Connection credentials are encrypted with tenant-bound authenticated encryption and stored with a key version so supported rotation can preserve the connection. Administrative summaries show identifying metadata rather than the secret itself. Company message policies govern editing, deletion and retention, while file and real-time access use their own checks.

Deployment configuration, provider choices and organizational operating procedures complete the security design. Identity, permissions, tenant boundaries and retained evidence give the organization concrete controls to examine during its own security review.

SECURITY, IDENTITY AND ACCESS

Security, identity and access / continued

Workflow and evidence

Organizational identity	Connect supported enterprise sign-in and directory administration through WorkOS, with the organization's configured identity setup and application entitlements.
Explicit application authority	Control sensitive operations through member, administrator, operator and customer policies. Role privileges and CSRF checks protect the applicable actions.
Company data boundaries	Resolve company scope before tenant data access and apply database row policies to company records. Operational lab views have a separate purpose from authorization.
Protected connection credentials	Store integration secrets with authenticated encryption, tenant-bound additional data and key versions that support rotation. Expose connection summaries without returning plaintext credentials.
Accountable business records	Retain the actor, revision and workflow context behind consequential decisions, including the controlled technical review and release lifecycle.
A clear assurance statement	Request deployment-specific evidence, operating responsibilities and any available independent assurance reports. Architectural controls alone do not establish a certification.

The customer relationship has its own access boundary.

A portal contact receives the records allowed for that customer and its permitted sites. Customer-service staff reach a conversation through the line's current responsibility rules, while internal team channels use their own membership rules. The application checks access at the relevant record and communication boundary.

SECURITY, IDENTITY AND ACCESS

Security, identity and access / continued

A lab, regional or company view helps a person focus operational information. It does not replace the permissions that determine what that person may read or do.

Security review should examine the actual deployment.

The deployment determines the identity provider, connection credentials, model endpoints and external services in use. An organization with strict data-control requirements can review those flows alongside application permissions, retention and operational controls.

A local model can address one part of that design. Mailbox providers, voice and video services, identity, monitoring and backups each retain their own responsibilities and data boundaries. No model choice supplies automatic compliance.

Related website guide: metraslate.com/platform/security

INTEGRATIONS AND TECHNICAL ARCHITECTURE

Define the Connection. Verify the Exchange.

Evaluate each integration by its supported operations, configuration and demonstrated outcome. A provider name alone does not establish a working connection.

Pilot / Early Access. Provider-dependent and subject to configuration and acceptance. No partner endorsement, universal connector or ready QuickBooks integration is claimed.

MetraSlate evaluates integrations against the actual workflow: which records move, which system owns them, how failures appear and what confirms the result. Provider registration, permissions and deployment configuration form part of that evaluation.

Mailbox, identity and calling adapters exist in the implementation. Their presence does not establish an active connection for your organization. Ask for the specific supported operation and an accepted provider demonstration before including it in a rollout.

Workflow and evidence

Email	Gmail API, Microsoft Graph and IMAP/SMTP transports have implementation paths. Provider access, permitted operations and live delivery require configuration and verification.
Identity	WorkOS-backed identity is configuration-dependent. Confirm your identity provider, membership model and SSO requirements.
Calling	RingCentral has an embedded-provider path. Registration, authorization and live call acceptance remain required.
Accounting	A ready-to-use QuickBooks connector is not offered here. Accounting exchange, posting authority and reconciliation require a separately agreed scope.

INTEGRATIONS AND TECHNICAL ARCHITECTURE

Integrations and technical architecture / continued

Measurement Exchange	An ingestion foundation exists. A specific executor, versioned mapping, supported instrument and end-to-end acceptance are still needed.
Payment Providers	Subscription billing infrastructure does not establish laboratory customer payment processing. See the billing page for the current boundary.

Start With the Data and the Outcome

Identify records, ownership, units, stable identifiers and the evidence each system must retain. Agree what happens when data is missing, refused, repeated or corrected.

Measurement Integration Requires Its Own Acceptance

External execution must be bound to the correct instrument, work context and controlled definition. Standard performs its supported uncertainty evaluation on accepted observations.

A complete local launch, cancellation, recovery and result-return workflow is not a universal shipped connector. Validate the selected execution system and mapping before depending on it.

Ask for a Scoped Integration Demonstration

The evaluation should establish the provider, operations, authentication, permissions, mapping, reconciliation and failure handling. An integration remains conditional until those checks succeed in the intended environment.

Related website guide: metraslate.com/platform/integrations

ENGINEERING AND TESTING

Standard is engineered and tested throughout development.

Numerical reference cases, controlled-record checks, permission refusals, database integration tests and browser workflows accompany the product from calculation kernels through release preparation.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

Standard's engineering follows the work a laboratory needs to trust: the number calculated, the evidence retained, the person allowed to act and the outcome recorded when a workflow continues or refuses. Automated checks address those responsibilities at several layers, with focused regression cases for the defects and boundary conditions found during development.

The numerical engine has reference cases with explicit expected values. Domain tests exercise approvals, corrections and failures. Database integration tests inspect persisted outcomes and company boundaries. Interface checks exercise record changes and browser behavior, while build, migration and release checks evaluate the candidate and its packaging.

Workflow and evidence

Reference values test the mathematics.	Check distribution divisors, uncertainty combination, effective degrees of freedom, Student-t coverage, decision boundaries and reporting precision against explicit expected results.
Units preserve the meaning of the quantity.	Exercise absolute values, uncertainty magnitudes, temperature offsets, electrical scaling and supported logarithmic conversions.
Historical evidence retains its calculation basis.	Check pinned definitions, correction scope, reviewed content, document hashes and rendering-version identity.
Refusal paths receive explicit tests.	Exercise missing authority, wrong account or site, invalid scope, absent evidence and operations that cannot safely continue.

ENGINEERING AND TESTING

Engineering and testing / continued

Transactions retain one coherent outcome.	Test receipt allocation rollback, statement-import deduplication, financial settlement events and repeatable document issuance.
Workflow and release checks complete the layers.	Connect backend and browser behavior with field-work reconciliation, migration validation, artifact checks and an explicitly reviewed release candidate.
Complete journeys with the relevant roles	Check connected customer and laboratory workflows using representative records, persisted outcomes and the permissions appropriate to each responsibility.
Backup and usable restoration	Exercise recovery of the database, documents and relationships, then inspect whether the recovered application can use the restored evidence.
Operational checks and actionable alerts	Connect service health, processing failures and integrity signals with the affected workflow, responsible person and recovery action.
Billing and reconciliation	Check charges, invoices, receipts, credits, balances, repeated input and reversals as one connected business workflow.

The calculation engine is checked against concrete numerical references.

The reference corpus includes a first-order, uncorrelated GUM end-gauge example. Its component values combine to a reported standard uncertainty of 32 nm; effective degrees of freedom select row 16, the 99% coverage factor is 2.9208 and the expanded result is 93 nm. The test retains those intermediate expectations so a changed result has a traceable place to investigate.

Further cases exercise rectangular, triangular and U-shaped distribution divisors, Welch-Satterthwaite effective degrees of freedom, Student-t table rows, guardband boundaries and TUR. Calculation precision and the reported number receive separate assertions, preserving the distinction between the working result and its displayed precision.

ENGINEERING AND TESTING

Engineering and testing / continued

Unit conversions are tested according to what the quantity represents.

The conversion cases include 330 mV to 0.330 V, 212 °F to 100 °C and a 0.5 °C uncertainty magnitude to 0.9 °F. Absolute temperatures carry their offset; uncertainty magnitudes use scale. Exact decimal factors and explicitly bounded numerical tolerances make the expected behavior reviewable.

Supported logarithmic conversions also have reference and round-trip cases. Unsupported logarithmic delta conversions receive an explicit refusal. These checks help preserve the meaning of readings, tolerance contributions and uncertainty inputs as records move between supported units.

CMC calculations and reporting boundaries have their own regression cases.

The CMC budget tests use an exact 3-4-5 contributor pair to check combination, expansion and the fitted gain-plus-offset line. Other cases exercise a contributor's stated coverage factor, Student-t selection, upward rounding of published CMC values and comparisons between claims and supporting budgets.

The reported-result floor has separate tests around the value printed on the certificate. A low-degrees-of-freedom result, a rounding boundary, an absent scope entry and the service policy's warning or refusal setting each receive their own case. This connects the numerical comparison to the actual operating decision.

Corrections and certificate review preserve the evidence behind an issued result.

A correction is checked against the predecessor's pinned point set and retained specification basis. Regression cases exercise missing points, repeated points, unknown points and later catalog changes. The resulting record can explain which definition and evidence supported its calculation.

Certificate checks examine the relationship between reviewed content and the document being sealed. A changed template or output version invalidates the earlier approval, and integrity cases deliberately alter stored evidence to check that the discrepancy is detected. A repeat issuance using the same issuance identity returns the existing capture.

ENGINEERING AND TESTING

Engineering and testing / continued

Company boundaries and permissions are tested through allowed and refused operations.

Tenant tests exercise database connections without a company context, transaction-local company stamps and reuse after a transaction ends. Database integration cases run competing work for two tenants and inspect which tenant records each operation can actually read.

Permission tests exercise anonymous access, absent write policies, missing privileges and stale or absent request tokens. Portal checks test both directions of the boundary between a customer's account and the laboratory workspace. The benefit is a reviewable set of checks around who can see a record and who can change it.

Financial and field workflows are checked for partial outcomes and repeated input.

Invoice tests inspect the balance derived from receipts and credits, the event emitted when that balance settles and refusal of an excessive allocation. A batch-receipt regression checks that an invalid allocation leaves the other invoices unchanged. Statement-import tests exercise identical files under different names and overlapping bank transactions.

Disconnected-work checks cross the browser sender and backend ingestion path, then inspect the readings and completed event retained after reconciliation. Scoped field credentials also receive refusal cases outside their authorized operations. These checks address the handoff between local work, server decisions and persisted evidence.

Browser, migration and release checks accompany the functional tests.

Interface regressions exercise behavior such as moving between quotations without carrying one record's PO information into another approval. Selected component and workflow stories run in a browser. Type checking, linting and production builds provide additional checks around the interface delivered to users.

The configured build pipeline also validates database migrations, tenant-policy state and document output. Its PDF/A check includes an expected conforming document and a deliberately invalid document so the validator must demonstrate both outcomes. Release preparation checks the packaged artifact, and the deployment workflow requires the exact reviewed commit.

ENGINEERING AND TESTING

Engineering and testing / continued

Verify the release through complete customer and laboratory journeys.

Release checks follow the connected work: a customer request, quotation and decision; receipt and service; measurement evaluation; technical review and released documents; and the financial and customer follow-through. Representative linked records and the relevant employee, reviewer, administrator and customer roles make those checks operationally meaningful.

Failure and recovery belong in the same journeys. Repeated requests, wrong-account access, interrupted processing, partial results, expired authority and corrected records are exercised with their persisted outcomes. The release record identifies the candidate, configuration and checks behind the decision to deploy.

Make backup and restoration part of release engineering.

Backup and restore procedures cover the database and the files needed to recover its records, documents and relationships. Restoration checks inspect the recovered information and the application's ability to use it, so recovery is evaluated against usable product data.

Release preparation also checks migrations, packaged artifacts and the path back from an unsuccessful rollout. The recorded revision and recovery procedure give the operator a concrete starting point when a deployment needs to be stopped or restored.

Expose operating health and turn failures into a response.

Service checks examine the application and the dependencies needed by its workflows. Failed processing, unavailable connections, delayed jobs and record-integrity problems produce identifiable operating signals with enough context for the responsible person to investigate.

Alerts connect the condition with the affected workflow and its recovery action. Staff can distinguish a service being reachable from a message being delivered, a result being accepted or a document being released, and can inspect the actual outcome before retrying an uncertain operation.

ENGINEERING AND TESTING

Engineering and testing / continued

Check billing as a connected business workflow.

Billing checks follow the relationship between the approved charge, invoice, receipt, credit, account balance and any external processing outcome. Partial settlement, repeated provider input, failed allocation and reversal are checked against the money applied to the actual obligation.

Customer-facing account information and laboratory records must tell a coherent story after those operations. Release evidence includes the normal path and recovery cases, helping the operator identify whether the next action is collection, reconciliation, customer follow-up or correction of the underlying record.

Related website guide: metraslate.com/platform/engineering

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and uncertainty budgets give a capability claim its numerical basis.

Create scope entries and contributor budgets, calculate supported endpoints, and publish a reusable technical curve. Keep numerical support separate from accreditation approval.

Pilot / Early Access. CMC budgets and technical curve publication are part of the Metrology pilot discussion. Broader scope lifecycle and accreditation treatment remain in validation.

Standard brings CMC budget calculation, scope entries and measurement uncertainty into the calibration workflow. A technical manager can maintain the evidence behind a claimed range while technicians and reviewers inspect the basis used for individual measurements.

The work begins with a defined reference standard and measurement function. The laboratory records the source or measure role, range, unit, optional frequency band, and claimed gain, offset and coverage factor. This gives the capability a numerical definition that can be checked against its supporting budget.

Workflow and evidence

The scope entry identifies what is claimed.	Reference-standard equipment, function, role, range, units and frequency qualification keep the claim attached to the capability it describes.
Each contributor retains its assumptions.	Low and high values, sensitivity, distribution, input coverage, degrees of freedom and exclusion state explain how the budget was assembled.
The endpoints support a reusable curve.	Standard calculates the endpoint budgets and fits a gain-plus-offset uncertainty curve for the declared band.
The claim can be checked against its support.	Claim states distinguish an uncomputed budget, a claim supported by its budget and a claim that is tighter than the configured evidence supports.

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and scope budgets / continued

Publication creates a controlled specification release.	The published curve carries its budget reference and coverage factor. Republishing replaces that budget's previous curve within one release.
The laboratory keeps responsibility for accreditation.	Standard organizes and calculates the supporting evidence. The laboratory and its accrediting body approve the scope and accreditation.

The contributor budget explains the capability.

A CMC budget retains contributor values at a low and a high point. Sensitivity and the selected distribution convert those inputs into standard uncertainty contributions. Supported inputs include normal-expanded, rectangular, triangular, U-shaped and standard-deviation values. An expanded-uncertainty input retains the coverage factor at which it was stated.

Standard combines the uncorrelated contributions by root-sum-square and calculates effective degrees of freedom. The scope-budget calculation can use a fixed coverage factor or its Student-t coverage basis. It applies one coverage factor across the two endpoints and rounds the expanded endpoint uncertainties upward to the supported publication precision before fitting the curve.

The claim and its supporting calculation remain connected.

A capability claim can become unsupported when a contributor changes. Standard distinguishes a claim that has no computed budget from one supported by the calculation and one that undercuts its supporting uncertainty. A refusal identifies the numerical disagreement so the laboratory can review the claim or its contributors.

Publishing places the fitted curve into the reference standard’s specification catalog. The curve retains its coverage and budget relationship. This is a technical publication; it is not an accrediting body’s approval of the laboratory’s scope.

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and scope budgets / continued

A built-in CMC calculator makes the arithmetic visible.

The engineering-calculator library also includes a CMC Uncertainty Budget tool. Its survey, low-point budget, high-point budget and equation views let the user examine contributor values, sensitivity, distributions, degrees of freedom, endpoint uncertainty and the fitted gain and offset.

Calculator drafts and saved results are stored locally in the user's browser. A calculator study is useful for examining the numbers; the scope-entry budget and controlled publication keep the laboratory's reusable capability definition attached to its reference standard.

Capability work continues into measurement review.

A published reference-standard curve supplies a contribution to later point evaluation. The calibration record then combines the actual attributed standards with the DUT-resolution contribution and the selected measurement policy. This connects capability preparation to the uncertainty reported for the work.

The laboratory can revisit contributors and ranges as its standards and methods change, while reviewing completed results against their retained calculation evidence. Budget authoring supports preparation and maintenance of scope evidence; it does not automatically submit an accrediting body's complete scope document.

Scope Review and Accreditation Boundaries

Scope submission, review and publication controls are in development and validation. The technical budget and the laboratory’s external accreditation have distinct purposes.

Effective-date resolution across completed results, full-range quantitative validation and result-level accredited treatment on mixed certificates require further validation. Confirm these requirements explicitly in the pilot; a computed endpoint or published curve does not establish them.

NIST TN 1297, Appendix B: Coverage Factors
Coverage-factor procedure and its assumptions, rather than a universal equivalence between a fixed k and a confidence level. 1994 edition, web section accessed 2026-10-21

BUILT-IN CMC AND SCOPE BUDGETS

Built-in CMC and scope budgets / continued

ILAC P14:09/2020: Measurement Uncertainty in Calibration
Authoritative policy source for calibration uncertainty and CMC context. P14:09/2020, official policy register accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/cmc

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty budgets keep the reported result explainable.

Follow the actual standard contributions, resolution, combination, coverage and decision basis from the recorded observation to the reported uncertainty.

Pilot / Early Access. Supported point uncertainty is demonstrated in Standard by MetraSlate: Metrology Edition. Repeated-observation completion and general correlated-input propagation remain outside the validated scope described here.

Standard performs the uncertainty evaluation for its calibration record. It resolves the contributions appropriate to the measurement, adds the configured DUT-resolution contribution and calculates the result under the selected service policy. The technician and reviewer can inspect the calculation at the point where it is used.

The result retains more than a final uncertainty number. Contributions, units, sensitivity, specification identity, coverage, TUR, guardband, decision outcome and reporting values provide the evidence needed to understand how the laboratory reached its conclusion.

Workflow and evidence

The attributed standards determine the contribution basis.	Standard resolves the relevant specification and retains the curve, units, operating coordinate, distribution or divisor, sensitivity and resulting contribution.
Resolution is part of the point budget.	The configured DUT resolution adds its contribution through a rectangular, triangular or explicitly configured divisor treatment.
Independent Contributions and Coverage	Combine supported independent contributions with their sensitivities. Inspect coverage, decision policy and reporting precision in the point result.
The reported value retains its calculation basis.	Full calculated expanded uncertainty and the reported uncertainty remain distinct, so reporting precision does not replace the underlying result.

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty evaluation / continued

The decision uses the laboratory's configured policy.	Coverage, TUR basis, guardband, acceptance limits, gates and the optional CMC-floor check form part of the evaluation context.
Completion preserves the evidence used.	The finished record keeps observations, component values, specification snapshots and decision evidence together for review and certificate lineage.

A reviewer can trace each contribution.

A reference standard's contribution begins with its attributed identity and the specification applicable to the point. The retained trace shows the authored curve, the operating value and unit, coverage or distribution treatment, sensitivity and the contribution carried into the DUT quantity.

When the standard measures a different quantity, the declared measurement relationship supplies the transfer. A missing relationship or missing specification produces a named refusal. The evaluator does not present an incomplete contribution set as a complete uncertainty budget.

The arithmetic can be inspected at the measurement point.

The point inspector separates setup from calculation. Its calculation view shows reference-standard contributions, process uncertainty, the DUT-resolution contribution, combined standard uncertainty, coverage and reported expanded uncertainty. The same view places limits, guardband, TUR and the decision basis alongside the calculation.

For example, independent standard uncertainties of 1.20, 0.80 and 0.60 microvolts combine to approximately 1.56 microvolts. With fixed $k = 2$, the expanded uncertainty is approximately 3.12 microvolts before the selected reporting convention. The laboratory approves the contributors, sensitivity assumptions and coverage appropriate to its method.

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty evaluation / continued

The measurement and its decision retain their policy.

Service policy controls how the supported uncertainty model reports its result and how the decision uses it. Standard retains the coverage factor, coverage probability, TUR basis, guardband, acceptance interval and the resulting outcome rather than collapsing them into a single pass flag.

The optional CMC-floor check compares reported uncertainty with a matching working-scope claim. It is not a complete accreditation determination or proof of effective-date treatment throughout a certificate.

Completed calculations preserve the basis used at the time.

Completion retains the raw and evaluated readings, contribution rows, calculation values, policy results and selected specification snapshots with the event. Later reference-data changes therefore do not silently replace the evidence behind the completed result. Historical records can also identify calculation detail that was not recorded at the time.

External observations require an agreed and validated ingestion contract before they can participate in this workflow. A measurement integration must preserve source identity and units; no universal instrument connector is implied.

Repeated Readings and Correlated Inputs

Repeated-reading summaries and statistical reference material are distinct from completed calibration uncertainty. Observation-set calculation and completion, general covariance propagation and uncertainty of an estimated sample spread are not included as validated pilot capabilities.

If your method depends on repeated observations or shared effects, bring the method and acceptance cases to the technical evaluation. An ordinary spread statistic must not be presented as its uncertainty.

NIST TN 1297, Section 3: Type A Evaluation of Standard Uncertainty
Statistical evaluation from observations and the reported quantity. 1994 edition, web section accessed 2026-10-21

BUILT-IN UNCERTAINTY EVALUATION

Built-in uncertainty evaluation / continued

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

NIST TN 1297, Appendix B: Coverage Factors
Coverage-factor procedure and its assumptions, rather than a universal equivalence between a fixed k and a confidence level. 1994 edition, web section accessed 2026-10-21

ILAC P14:09/2020: Measurement Uncertainty in Calibration
Authoritative policy source for calibration uncertainty and CMC context. P14:09/2020, official policy register accessed 2026-10-21

JCGM 106:2012: The Role of Measurement Uncertainty in Conformity Assessment
Decision-rule and conformity-assessment reference alongside the declared measurement model. JCGM 106:2012, official publication page accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/uncertainty

CONDITIONAL SPECIFICATIONS AND INSTANCE CORRECTIONS

Evaluate the specification that applies to the instrument's operating conditions.

Connect temperature, elapsed time, warm-up, mode, frequency, loading and recorded environmental conditions to controlled specifications and instrument-specific corrections.

Pilot / Early Access. Confirm the enabled workflow, roles and configuration in the scoped pilot.

An accuracy statement is meaningful within its stated operating conditions. Standard retains those conditions with the model, selects the applicable specification for the point and exposes the calculation used to turn that specification into a contribution to the measurement result.

Manufacturer limits, instance calibration corrections, environmental influences and DUT acceptance limits have different roles. Keeping their provenance, validity and uncertainty explicit helps the laboratory apply the right information without silently substituting one for another.

Workflow and evidence

Temperature terms keep their defined reference.	Retain the base accuracy, reference temperature or band, valid temperature limits, coefficient form and the recorded temperature used in the evaluation.
Operating conditions determine applicability.	Use time since calibration, warm-up, autocalibration, resolution, mode, frequency, loading and other stated conditions to resolve the specification that governs the point.
Environmental observations retain their context.	Record conditions by run or point with source, timestamp and units, and identify missing, changing or unsuitable observations.
Instance corrections retain their certificate evidence.	Keep instrument-specific coefficients and certificate-derived corrections separate from manufacturer limits, with validity, uncertainty, conditions and revision.

CONDITIONAL SPECIFICATIONS AND INSTANCE CORRECTIONS

Conditional specifications and instance corrections / continued

Piecewise boundaries resolve deterministically.	Make overlaps, gaps, endpoint inclusion and precedence explicit so the same inputs select the same applicable rule.
Draft recomputation preserves released history.	Changes invalidate dependent draft results and trigger recomputation, while released evidence continues through controlled amendment and lineage.

Temperature-dependent accuracy retains its formula and valid range.

A temperature model records the base accuracy, reference temperature or reference band, valid temperature limits and coefficient form. The author can express supported terms as ppm, a percentage of reading or range, an absolute quantity or an accuracy multiplier, with the applicable function and range identified.

The evaluation retains the observed temperature, its source and time, the coefficient, the resulting contribution and the specification revision. Calibration temperature and autocalibration temperature are separate conditions where the authored specification uses them. The model identifies how each term applies and avoids counting a temperature allowance again when it is already included in the base statement.

A temperature allowance remains an accuracy or uncertainty term unless an authored correction model defines a signed adjustment. It does not silently change the DUT acceptance limit, and the resolver does not extend the coefficient beyond its declared conditions.

Elapsed time and instrument state distinguish a modifier from a requirement.

Conditional specifications can depend on time since calibration, warm-up duration, autocalibration status, resolution or digits, operating mode, frequency, loading and environmental conditions. A condition may select a different base statement, contribute a numerical modifier or define when a specification is valid.

Standard preserves that distinction. A missing warm-up requirement is not converted into an arbitrary extra uncertainty, and an operating-mode restriction is not treated as a coefficient. The reviewer can inspect the conditions that selected the rule, the modifiers applied and any condition that prevents the specification from being used.

CONDITIONAL SPECIFICATIONS AND INSTANCE CORRECTIONS

Conditional specifications and instance corrections / continued

Environmental observations and instance corrections retain their provenance.

Environmental values can apply to the run or to an individual point. Each observation retains its source, capture time and units, allowing a changing condition to be related to the point measured at that time. Missing, stale or unsuitable values remain identifiable rather than being replaced by one assumed ambient value for the entire job.

An individual instrument's calibration coefficients or certificate-derived corrections retain their own source, validity period, uncertainty, operating conditions and revision. The measurement model states how the correction applies to the result and how its uncertainty contributes. Manufacturer performance limits stay separately identifiable, so an instance correction cannot silently replace a limit or become a second copy of an already included contribution.

Deterministic rule selection and dependency tracking protect the result.

Piecewise specifications identify their ranges and the inclusion of each boundary. Selection rules make overlaps, gaps and precedence explicit, and the same instrument, point and conditions resolve to the same rule. A gap or an incompatible condition receives a named outcome instead of a guessed coefficient or hidden extrapolation.

A change to readings, conditions, specification revisions, corrections or calculation settings invalidates the draft results that depend on it and recomputes their analysis. The reviewer can inspect the inputs and rule set used for the new result before release.

Released records preserve their evaluated evidence. A later correction proceeds through controlled amendment with predecessor and successor lineage, keeping the earlier result and the reason for the change available. The laboratory can improve a model without rewriting the history of work already released.

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

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Related website guide: metraslate.com/metrology/specifications

MEASUREMENT ANALYSIS AND STUDY PLANNING

Understand the Calculation Before You Rely on the Result.

Explore the calculator and statistical reference tools, with the distinction between supported calculations and planned measurement-system studies made explicit.

Pilot / Early Access. Calculator and reference tools are available for demonstration. Complete MSA study workflows are planned.

The calculator library includes CMC, mathematical and statistical tools. The selected tool's inputs, assumptions and persistence determine what its result can support.

Full Gage R&R, bias, linearity, stability and attribute-agreement study workflows are planned. Their reference material does not establish a complete study workspace, report or customer-available analysis path.

Workflow and evidence

CMC Budgets	Inspect contributor inputs, coverage and the resulting capability calculation.
Statistical Tools	Confirm the specific calculator, input format and saved-result behavior in your demonstration.
Study Planning	Use the references to define the study and its acceptance criteria. Full MSA workflows remain planned.
Separate Quantities	A mean, sample standard deviation and uncertainty answer different questions. Check the supported output before applying it.

Calculators and Retained Records

Calculator drafts and saved results may live in the browser. A calculator result is distinct from the controlled laboratory record and its completed uncertainty evidence.

MEASUREMENT ANALYSIS AND STUDY PLANNING

Measurement Analysis and Study Planning / continued

A Study Needs More Than a Formula

Parts, operators, trials, methods, observations and reporting must form one accepted workflow before a complete MSA capability can be offered. Discuss the particular study you need as a future requirement; do not assume it is included in today's pilot.

NIST TN 1297, Section 3: Type A Evaluation of Standard Uncertainty
Statistical evaluation from observations and the reported quantity. 1994 edition, web section accessed 2026-10-21

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

NIST/SEMATECH Engineering Statistics Handbook, Section 2.4: Gauge R&R Studies
Study design, data collection, repeatability, reproducibility, stability and bias investigations. Chapter 2 publication 2003, online section accessed 2026-10-21

JCGM 106:2012: The Role of Measurement Uncertainty in Conformity Assessment
Decision-rule and conformity-assessment reference alongside the declared measurement model. JCGM 106:2012, official publication page accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/measurement-analysis

CALCULATION AND PROCESS REFERENCE

Every calculation has a purpose, a model and a reference.

Inspect the intended use, inputs, units, assumptions, formula, numerical example and limits of a calculation before applying its result to the laboratory's work.

Pilot / Early Access. Method guidance includes topics beyond current calculation and completion support. The uncertainty and measurement-analysis pages state the current boundaries.

Standard places a calculation and process reference beside its measurement tools. Each entry identifies what the calculation is for, the quantity it reports, its required inputs and conditions, and the mathematical operation it performs. A worked numerical chain lets the reader follow the inputs through intermediate values to the reported output.

The reference library covers supported calculations and planned methods. Each reference is explanatory guidance, not evidence that every described method is implemented or available in the pilot. Match it to the actual tool and retained result shown in your evaluation.

A laboratory can review the method before configuring a test, then open the same reference while inspecting a result or investigating a changed number. Formula, units and validity remain visible together, reducing the risk that a familiar statistic is applied to a different reported quantity or beyond the conditions the method supports.

Workflow and evidence

Purpose identifies the question the tool answers.	Read the intended use and the reported quantity, including the distinction between a mean, a sample standard deviation, mathematical RSS and measurement uncertainty.
Inputs retain their meaning and origin.	Inspect the required quantities, units, raw observations, source documents and evidence before accepting the calculation.
Assumptions define the valid application.	Review independence or covariance, statistical conditions, required sample structure and the operating range for which the selected method applies.

CALCULATION AND PROCESS REFERENCE

Calculation and process reference / continued

Formulas expose the calculation chain.	Follow the declared equation, sensitivity, combination, coverage and reporting operations with their intermediate values.
Numerical cases make the method reviewable.	Compare a worked input-to-output chain with the selected calculation version and its declared precision and rounding boundaries.
Authoritative references retain their edition.	Open the relevant NIST, JCGM or calibration-policy reference and identify the source version used by the method entry.
Guidance and result evidence keep separate roles.	The reference explains the calculation. The result retains the model revision, inputs and evidence that were applied to the work.
Limits prevent silent changes of meaning.	Check the method's conditions and named refusals before using a result as an uncertainty, capability claim, study conclusion or conformity decision.

NIST TN 1297, Section 3: Type A Evaluation of Standard Uncertainty
Statistical evaluation from observations and the reported quantity. 1994 edition, web section accessed 2026-10-21

NIST TN 1297, Appendix A: Law of Propagation of Uncertainty
Measurement model, sensitivity, covariance and the standard uncertainty of a mean under the stated independent-observation conditions. 1994 edition, equations A-1 to A-5, web section accessed 2026-10-21

NIST TN 1297, Appendix B: Coverage Factors
Coverage-factor procedure and its assumptions, rather than a universal equivalence between a fixed k and a confidence level. 1994 edition, web section accessed 2026-10-21

NIST/SEMATECH Engineering Statistics Handbook, Section 2.4: Gauge R&R Studies
Study design, data collection, repeatability, reproducibility, stability and bias investigations. Chapter 2 publication 2003, online section accessed 2026-10-21

CALCULATION AND PROCESS REFERENCE

Calculation and process reference / continued

ILAC P14:09/2020: Measurement Uncertainty in Calibration
Authoritative policy source for calibration uncertainty and CMC context. P14:09/2020, official policy register accessed 2026-10-21

JCGM 106:2012: The Role of Measurement Uncertainty in Conformity Assessment
Decision-rule and conformity-assessment reference alongside the declared measurement model. JCGM 106:2012, official publication page accessed 2026-10-21

These references explain methods and their conditions. They do not establish that every described method is implemented, validated or available in the pilot.

Related website guide: metraslate.com/metrology/calculation-reference

OPTIONAL AI AND PRIVATE MODELS

AI is not required. It is optional.

Run the operation through controlled records, calculations and workflows. Add model assistance where it helps, with a supported model and deployment approach suited to your data-control requirements.

Pilot / Early Access. Optional and configuration-dependent. Confirm the model, information sent and hosting boundary before enabling assistance.

Standard's working foundation is its records, deterministic workflows, calculations, approvals and business controls. Equipment identity, customer requests, work progress, uncertainty, technical review and document release do not depend on a model answer. Organizations can adopt the platform on the strength of those capabilities and choose where optional assistance is useful.

Conversational assistance can help people work through a question in the application. The configured model returns a streamed answer, and the completed exchange retains which model answered and its reported usage. Attached record references identify the records named in a discussion; they do not give the model unrestricted access to the database or automatically authorize a business action.

A separate optional workflow can identify commitments in selected communication. It requests a defined result, checks the response against the extraction schema and retains its relationship to the source message. An unresolved person or a result below the configured confidence threshold waits for review. Eligible results can enter the commitment workflow with their source retained.

Supported OpenAI-compatible endpoints allow an appropriate local or privately operated model to participate in the architecture. Model and endpoint selection are configuration choices, with different arrangements for shared hosting and standalone deployments that can reach private-network services. This gives organizations with strict data-control requirements a concrete deployment question to resolve before enabling assistance.

The data boundary must include the text actually sent to the selected endpoint. Commitment extraction uses participant tokens, but the selected message body remains part of the request. Conversational assistance sends the exchange needed to answer. Organizations should select the model, endpoint and use cases according to the information they permit that service to process.

Private model choice addresses the model connection. Identity, mail, maps, calling, video, monitoring and backups each have their own configuration and service boundaries. No model automatically guarantees compliance, and a private model does not make every connected service private.

OPTIONAL AI AND PRIVATE MODELS

Optional AI and private models / continued

Workflow and evidence

A complete deterministic foundation	Use controlled records, calculation logic and workflow decisions without requiring AI. Optional assistance adds to the operation rather than becoming a prerequisite for it.
Conversational assistance	Ask a question in context and retain the completed exchange with its model identity and reported usage. Referenced record names do not imply unrestricted record-content access.
Commitments with source context	Identify candidate commitments through a defined extraction workflow. Schema checks, source spans and configured review thresholds preserve an inspectable relationship to the original message.
Supported local and private models	Use an appropriate OpenAI-compatible endpoint, including supported locally operated runtimes. Reachability, hosting posture and the organization's approved network boundary determine the deployment.
Bounded provider access	Configured extraction calls use connection budgets, response limits, concurrency controls and endpoint-address policy. An unavailable or invalid provider result remains a visible outcome.
Strict data-control deployments	Agree the model endpoint and permitted information, then examine the other connected services separately. Model selection supports a deployment design; it does not confer automatic compliance.

OPTIONAL AI AND PRIVATE MODELS

Optional AI and private models / continued

Choose assistance for a useful job.

The clearest starting points are bounded: help a person think through a question, or identify a commitment in communication they have selected for that purpose. The record and workflow supply the business context; the model supplies an answer or structured extraction that can be inspected.

Technical calculations and release authority retain their own deterministic and human responsibilities. Optional assistance does not replace the approved uncertainty method, a technician's signing responsibility or the laboratory's release decision.

Put the deployment boundary before the model name.

For an organization with strict data-control requirements, the first question is where the selected text may be processed. A private-network model needs a deployment that can reach it, while a shared hosted application has a different endpoint policy. A supported model protocol alone does not settle the hosting arrangement.

Related website guide: metraslate.com/platform/ai

BILLING AND PAYMENTS

Keep the Approved Work Connected to Its Invoice.

Retain the customer's agreement, connect the invoice to the accepted work, and follow recorded receipts and balances. Payment processing has a separate adoption boundary.

Pilot / Early Access. Invoice and receipt records can be demonstrated. Active payment processing and customer self-payment are not included; confirm the accepted-agreement controls for your pilot.

Quotations, customer decisions, invoices and recorded receipts provide the commercial record around a service. A purchase order or accepted quotation records an agreement; it does not mean money has been received.

Newer agreement controls retain the reviewed estimate revision, approved subset and terms. Their availability must be confirmed against the pilot configuration. Live collection, card holds, automatic charging, refunds and customer self-payment are not active in the offer described here.

Workflow and evidence

Quotation and Approval	Keep the agreed revision and customer decision attributable to the work.
Invoice and Balance	Connect billed work, recorded receipts and the amount remaining. A recorded receipt is distinct from a processor transaction.
Customer Access	Authorized customers can view the invoices and account information made available by their laboratory.
Processing Boundary	Agree the external payment process during pilot setup. Do not assume card processing or self-payment is included.

An Approval Has a Specific Basis

The quotation gives the customer a defined service and price to review. Revised work needs its own clear decision. Retaining the agreement helps staff explain what was approved when preparing the invoice.

BILLING AND PAYMENTS

Billing and payments / continued

The Invoice Explains the Account Position

Invoice lines, recorded receipts and balances belong with the customer and work they concern. Review access and the accounting process during setup so people know which system records each fact.

Payment Automation Is a Separate Development

Configurable funding requirements, card authorizations, provider collection, refunds and self-payment need additional activation and acceptance. They are not implied by the presence of an invoice, approval or payment record. Bring these requirements to the pilot discussion.

Related website guide: metraslate.com/platform/payments

TECHNICAL RECORD

Follow the result back to its basis.

A useful evaluation inspects what was recorded, what was applied and what remains retrievable after review.

Identity	Instrument, customer and site, work-order line, controlled method and datasheet revision.
Observation	Value and unit, stimulus or nominal, phase, adjustment, environment, attributed standards and capture context.
Evaluation	Specification identity, supported contributions, coverage, TUR, guardband and decision outcome. Distinguish full calculation values from reported values.
Release and correction	Exercise review, certificate release and authorized customer access. Confirm how correction lineage, retained evidence and earlier documents behave in the selected configuration.

Current scope

Repeated-observation completion, general covariance propagation and broader scope/accreditation treatment require the separate limits described in this guide.

INTEGRATION ACCEPTANCE

Preserve meaning across the exchange.

These are evaluation requirements for an agreed connection. They are not a claim that every external system already has an accepted connector.

Authority and identity	Name the record owner, stable execution and observation identifiers, instrument, work context, method revision, units and phase.
Provenance	Agree which payload evidence, timestamps, actor or station attribution, mapping version and processing receipts must be retained.
Recovery	Exercise missing data, refusals, repeated delivery, conflicting revisions and correction. Establish which system owns the final outcome.
Measurement	Compare equivalent models, quantities, units, contributors, coverage and rounding. External execution must be accepted against the selected mapping and supported Standard evaluation.
Accounting and providers	A ready QuickBooks connector is not offered here. Separately agree posting authority, ledger mapping, synchronization and reconciliation. Provider registration and live acceptance remain necessary for enabled connections.

CAPABILITY BOUNDARIES

Know what needs its own acceptance.

The pilot is the place to inspect both supported behavior and the requirements still in development or validation.

Built-in CMC and scope budgets	CMC budgets and technical curve publication are part of the Metrology pilot discussion. Broader scope lifecycle and accreditation treatment remain in validation.
Built-in uncertainty evaluation	Supported point uncertainty is demonstrated in Standard by MetraSlate: Metrology Edition. Repeated-observation completion and general correlated-input propagation remain outside the validated scope described here.
Measurement Analysis and Study Planning	Calculator and reference tools are available for demonstration. Complete MSA study workflows are planned.
Billing and payments	Invoice and receipt records can be demonstrated. Active payment processing and customer self-payment are not included; confirm the accepted-agreement controls for your pilot.
Integrations and technical architecture	Provider-dependent and subject to configuration and acceptance. No partner endorsement, universal connector or ready QuickBooks integration is claimed.
Analytics and Reporting	Reporting foundations can be demonstrated. Subjects, permissions and package exposure must be confirmed; a complete service-costing workflow is not claimed.

BUYING AND IMPLEMENTATION

Agree what the pilot must prove.

Bring an equipment list, representative records and a real workflow. Use the pilot to settle fit, responsibilities and the adoption decision.

Demonstrate the fit	Identify the product, edition, people, sites and workflows to evaluate. Name the customer acceptance owner and MetraSlate contact.
Agree commercial scope	Confirm enabled capabilities, licensing basis, quoted charges and inclusions. Identify separately charged implementation, import assistance, training or support. No standard price or rollout timeline is published here.
Validate a sample	Supply an authorized representative export. Current CSV and XLSX import creates new equipment records; full historical migration, relationships and attachments require separate review and acceptance.
Accept the workflows	Exercise the actual roles, permissions, record outcomes and failure cases in the pilot configuration. Keep unresolved requirements visible before wider adoption.
Complete the handover	Agree hosting and infrastructure ownership, updates, support contacts, backup operator and restore owner. Set retention and recovery expectations, then define export formats, limits, assistance and exit responsibilities.

Buying and implementation: metraslate.com/hosting

ASSURANCE REVIEW

Request evidence for your deployment.

Source and reference descriptions inform evaluation; the accepted configuration and operating evidence establish the customer behavior.

Security	Request deployment-specific access, tenant isolation, retention and recovery answers. This overview asserts no security certification, accreditation or penetration-test result.
Operations	Name application and infrastructure operators, monitoring and update responsibilities, backup and restore owners, support contacts and escalation.
Acceptance	Record the scenario, configuration, role, outcome and remaining limitation. A successful build, adapter name or screenshot alone does not establish a working customer integration.

A PRACTICAL EVALUATION

Bring the work you need to prove.

Request a demonstration built around the records, decisions and people that matter to your organization.

Equipment owners	Follow identity, location, service history and supporting files. Confirm whether focused AssetLight or the broader Asset business scope fits your needs.
Calibration laboratories	Follow a customer request into the work order, recorded observation, supported calculation, review and released certificate. Inspect the retained basis, not only the final number.
Technical and operating owners	Agree configuration, permissions, integrations, migration, support, recovery and export responsibilities. A provider name or recorded receipt does not prove a live connection or processed payment.

Request a demonstration at metraslate.com/demo